# VA-ATACOM Supplementary Material: The DT-margin Heuristic Baseline

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Table 1: DT-margin ablation on Goal (5 seeds).

| Config | $\alpha$ | $h$ | Mean $\bar{C}$ | GO/5 |
|---|---|---|---|---|
| Velocity-adaptive only | 0.3 | 0 | 2.90 | 5/5 |
| BRT-only | 0.0 | 3 | 2.35 | 4/5 |
| **DT-margin (full)** | 0.3 | 3 | **0.96** | **5/5** |

## The DT-margin Heuristic Baseline

VA-ATACOM's velocity augmentation has a heuristic predecessor we call the **DT-margin** filter — the "DT-margin (heuristic)" row of Table 1 in the main paper. It inflates the keepout radius linearly with speed, $r_{\text{eff}}(\boldsymbol{v}) = r_{\text{base}}(1+\alpha\|\boldsymbol{v}\|)$, projects out the inward radial component, and adds the same sim-BRT reward shaping. As shown in the main paper's design study, this linear margin is the *first-order* approximation of VA-ATACOM's quadratic braking constraint; under fair hazard-radius matching it reaches 13/15 GO — two Push seeds short of VA-ATACOM's 15/15, illustrating that the linearisation almost captures the right behaviour but leaves a residual gap that only the exact quadratic constraint closes. We characterise it here for completeness — its two tunable knobs ($\alpha$ and the BRT horizon $h$) and their interaction with the control rate. These findings motivated, and are superseded by, the parameter-free velocity-augmented manifold of the main paper.

### Ablation of the two components

Table 1 (Fig. 1) shows both components contribute: velocity-adaptive margin alone reaches 5/5 GO at higher variance (2.90); BRT alone 4/5 GO (one M1 collapse at $C$=7.98); the full heuristic 0.96 cost / 100% GO.

### $\Delta t$ sensitivity

Table 2 corroborates the M1 chord-excursion premise of the main paper's failure-mode analysis (and hence Prop. 4): continuous-time ATACOM achieves zero cost at $\Delta t \leq 0.02$s but fails at $\Delta t \geq 0.05$s, the phase transition matching the M1 threshold $\Delta t\|u\| \gtrsim \sqrt{d_{\text{safe}} \cdot r}$. The DT-margin heuristic tracks the safe regime at $\Delta t \in \{0.10, 0.20\}$s with $h$=3. At $\Delta t$=0.05s the picture is nuanced: the horizon rule prescribes $h^{\star}$=4, but a 5-seed sweep finds $h$=3 at 3/5 GO (bimodal), $h$=4 at 0/5, and $h$=5 at 3/5 (one seed at 37.05). We attribute

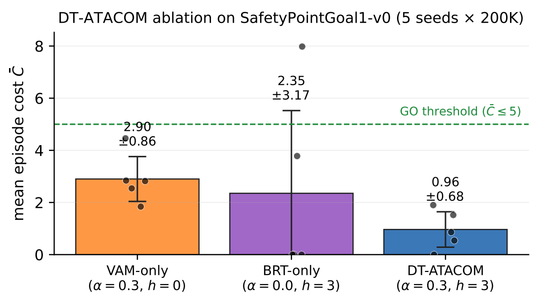


Figure 1: DT-margin ablation: both components are necessary. Mean episode cost $\bar{C}$ on SafetyPointGoal1-v0 (5 seeds $\times$ 200K env steps), bars are $\pm$1 std. VAM-only ($\alpha$=0.3, $h$=0) addresses M1 chord excursion only; BRT-only ($\alpha$=0, $h$=3) anticipates M2 drift but one seed collapses via M1; DT-margin ($\alpha$=0.3, $h$=3) combines both for lowest mean (0.96) and 5/5 GO.

Table 2: DT-margin cost vs. time step ($\bar{C} \pm \sigma$, 5 seeds, 50K steps).

| $\Delta t$ | ATACOM | DT-margin ($h$=3) | Note |
|---|---|---|---|
| 0.01s | 0.00 | 3.02 | ATACOM OK |
| 0.02s | 0.00 | 0.00 | ATACOM OK |
| 0.05s | 39.95 | 4.96±6.08 (3/5 GO) | Basin regime |
| 0.10s | 13.68 | **2.60** | DT-margin OK |
| 0.20s | — | 1.33±1.50 | $h \gg h^{\star}$ |

this to a *BRT-shaping basin*: near the penetration horizon the lookahead reports unavoidable penetration along multiple directions every step, integrating spurious penalty into the policy gradient and trapping training.

### $\alpha$/$h$ sensitivity

Sweeping $\alpha$ and $h$ on Goal (5 seeds $\times$ 200K):

Only configurations at least one step above the minimum on *both* axes ($\alpha$≥0.1, $h$≥2) reach 5/5 GO; setting $\alpha$=0 drops to 4/5 with a seed at $\bar{C}$=8.46. The over-provisioned config ($\alpha$=0.5, $h$=5) *regresses* to 4/5 ($\bar{C}$=9.26 outlier) — large-$h$ BRT shaping can integrate spurious penalty (see the $\Delta t$ sensitivity above) and destabilise training. The DT-margin recipe

Table 3: DT-margin $\alpha/h$ sensitivity on Goal (5 seeds $\times$ 200K).

| Config | $\alpha$ | $h$ | mean $\bar{C}$ | GO/5 |
|---|---|---|---|---|
| no margin | 0.0 | 2 | 3.37±2.81 | 4/5 |
| short horizon | 0.05 | 1 | 1.94±1.78 | 4/5 |
| both at minimum | 0.05 | 2 | 2.36±2.60 | 4/5 |
| above minimum | 0.1 | 2 | **1.30±1.66** | **5/5** |
| $\alpha$ generous | 0.3 | 1 | 1.72±1.46 | 5/5 |
| **paper choice** | 0.3 | 3 | **0.96±0.68** | **5/5** |
| over-provisioned | 0.5 | 5 | 3.00±3.38 | 4/5 |

is thus robust only with margin on both knobs; VA-ATACOM eliminates $\alpha$ entirely, replacing it with the calibrated $a_{\max}$.

## Computing Infrastructure

All experiments reported in the main paper and this supplementary were run on a single workstation:

- **CPU**: Intel Core i7-13850HX (20 cores, 28 threads).
- **RAM**: 31 GiB DDR5.
- **OS**: Ubuntu 24.04 LTS, Linux kernel 6.17.
- **GPU**: not used — all training and evaluation run *single-threaded on CPU* (`OMP_NUM_THREADS=1`, `MKL_NUM_THREADS=1`). Each Safety-Gymnasium 200K-step run takes ≈ 15–20 min wall-clock; we parallelise up to 20 such single-threaded runs at once.

Software stack (versions used at experiment time):

- Python 3.10
- PyTorch 2.x (CPU build)
- NumPy 1.23 / matplotlib 3.x
- `safety-gymnasium` 1.0 (the Safety-Gym successor; we ran with `SafetyPointGoal1-v0`, `SafetyPointPush1-v0`, `SafetyPointMultiGoal0-v0`)
- `gymnasium` 0.28 / `mujoco` 2.3
- Webots R2023b (for the rigid-body validation in main § Sim-to-Real)

Random seeds 0, 1, 2, 3, 4 are used for every (method, task) cell in Table 1 of the main paper. Total compute for the full Table 1 + Webots study is approximately **200 CPU-hours** on the workstation above ($\sim$ 10 h wall-clock with 20-way parallelism); the L-shape audit re-run added a further $\sim$ 5 CPU-hours.